Guillermo Fernando Farfán Romero

Senior Database Administrator (DBA) & Data Engineer

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PROFESSIONAL SUMMARY

Senior Database Administrator (DBA) and Data Engineer with 8+ years of experience architecting, administering, migrating, optimizing, and scaling mission-critical relational, analytical, NoSQL, and vector database platforms. Specialist in **Microsoft SQL Server (2012–2022)**, **PostgreSQL (with pgvector)**, **SAP HANA (IaaS & Backint ingestion)**, **MySQL, and MongoDB**, advanced **Performance Tuning** (execution plan analysis, table partitioning, indexing, deadlock resolution), cloud data architectures on **Google Cloud Platform (BigQuery, Cloud SQL, Cloud Storage)**, and automated ETL/ELT pipelines in **Python (SQLAlchemy 2.0, Pandas, Polars, PyArrow) and Apache Airflow**.

Proven track record leading enterprise database migrations to the cloud (**SQL Server, PostgreSQL, and BigQuery on GCP**), high-throughput storage optimization, and **DRP/BCP architectures with RPO < 15 min and RTO < 1 hour**.

TECHNICAL SKILLS & COMPETENCIES

- **Database Engines (RDBMS, In-Memory & Vector):** Microsoft SQL Server (2012–2022, AlwaysOn Availability Groups, T-SQL, SSIS/SSRS), PostgreSQL (12–16, pgvector, Declarative Partitioning, Streaming Replication, PgBouncer), MySQL, MariaDB.
- **Data Warehousing, Lakehouse, NoSQL & BI:** Google BigQuery (Partitioning, Clustering, Star Schema / Kimball, Private Service Connect - PSC), Redis (Distributed Caching, Pub/Sub, In-Memory Persistence), MongoDB, Google Cloud Pub/Sub, Power BI, Looker Studio.
- **Data Engineering & Automation:** Python (SQLAlchemy 2.0, Pandas, Polars, PyArrow, PyODBC), Apache Airflow, Alembic (Zero-Downtime Migrations), dbt, n8n, Baserow, Celery, PowerShell & Bash Scripting.
- **Cloud Infrastructure & Storage:** Google Cloud Platform (Cloud SQL, BigQuery, Cloud Storage, Cloud KMS), Terraform, Linux (Debian, Ubuntu, RHEL), XFS, LVM.
- **Diagnostics, Observability & Continuity:** Query Execution Plans, SQL Profiler, Extended Events, Index Fragmentation Management, Deadlock Resolution, OpenTelemetry (OTel), Prometheus, Grafana, Strict RPO/RTO Backup/Restore Strategies, RBAC, TDE/KMS/SSL Encryption.

PROFESSIONAL EXPERIENCE

Senior Python Backend & Financial Automation Engineer | Reforest Latam

May 2026 – Present | Full-time (Remote)

- Architected and developed end-to-end the financial operations platform and transactional Telegram bot in **Python 3.11+ (aiogram 3.x, asyncio, Pydantic)** with **Redis-persisted FSM** state management (RedisStorage and events_isolation double-click protection), shared HTTP connection pooling (aiohttp), exact decimal accounting (ROUND_HALF_UP), real-time **fail-closed RBAC**, and **Baserow / Google Drive API** integration.
- Built an **atomic multi-currency bank reconciliation engine** featuring native PDF/CSV/Excel statement parsers (BBVA, Galicia, Macro, Mercury, Balanz), **SHA-256** deduplication in Redis, decoupled staging with semantic auto-classification, batch injection with **atomic batch rollback**, historical **MEP Dollar** exchange rate lookup with T-1 to T-5 business-day fallback, and sanitized **ARCA** tax exports.
- Implemented **dual LLM OCR & automated classification (**Gemini 2.5 Flash **and** DeepSeek **via OpenRouter)** using semaphore-bounded concurrent PDF chunking and prompt-injection sanitization; deployed on **Linux VPS with Docker Compose**, automated daily Drive backups, and **GitHub Actions CI/CD** running **Ruff** and **514 passing Pytest integration tests (100% green)**.

Senior Data Infrastructure Consultant & DBA | Independent Consulting & Enterprise Projects

January 2021 – Present | Strategic B2B Technology Consultant (Independent / Contract / Remote)

- **Database Migrations & GCP Cloud Architectures (Enterprise Clients under NDA):** Designed and provisioned with **Terraform** high-availability database environments on **Google Cloud Platform**, private interconnection to **BigQuery Data Lakehouses**, and migrations of production **SQL Server 2017/2019 and PostgreSQL** engines with full transactional integrity.
- **ETL Pipelines, BI & Performance Tuning (Fintech & Industrial Clients under NDA):** Built asynchronous data pipelines and eliminated N+1 bottlenecks using **Python (SQLAlchemy 2.0, Pandas, Polars, PyArrow, Celery), Airflow, and BigQuery**, driving **>60% reductions in p95 query latency and batch processing windows**, alongside executive dashboards in **Power BI and Looker Studio**.
- **Database Observability & Operational Continuity (Clients under NDA):** Implemented deep database telemetry on SQL Server and PostgreSQL using **OpenTelemetry, Prometheus, and Grafana**, monitoring TPS and p95/p99 query latencies while cutting **MTTR by 45%**.

Senior Software Engineer & Database Administrator (DBA) | University of Buenos Aires (UBA)

March 2018 – February 2024 | Institutional Systems Specialist (Part-time / Buenos Aires, Argentina)

- Spearheaded High Availability (HA) administration and **Performance Tuning** (execution plans, index restructuring, deadlock resolution in T-SQL and PL/pgSQL) across **8+ TB of institutional data** on **SQL Server and PostgreSQL** serving **+50,000 users (+10,000 concurrent active users)**, cutting **p95 query latency by 55% (-40% average response time)**.
- Executed **zero-loss migration of +500 GB of critical legacy databases**, guaranteed **99.9% uptime (RPO < 15 min and RTO < 1 hour)** through DRP/BCP strategies, and automated ETL pipelines with **Python, Alembic, and Bash (-60% manual operational effort)**.

Systems Infrastructure & Operations Coordinator | Government of the City of Buenos Aires (GCBA)

January 2020 – December 2023 | Ministry of Economic Development (Full-time / Buenos Aires, Argentina)

- Administered, tuned, and monitored **SQL Server, PostgreSQL, MySQL, and MongoDB** databases for citizen portals serving **100,000+ monthly users (99.9% SLA)**, optimizing native compression and differential backup rotation (**-35% overnight backup windows and -95% data loss risk**), standardizing validation layers (**-40% data inconsistencies**), and training **+30 officials** in SQL.
- *Concurrent Public-Sector DBA Experience:* **Database Administrator & Data Analyst** at **Ombudsman's Office of the City of Buenos Aires (Defensoría del Pueblo CABA)** *(Jun 2018 – Jun 2022)*, administering **SQL Server and MySQL** and building **+100 institutional analytical reports**.

EDUCATION, CERTIFICATIONS & LANGUAGES

- **Education: Associate Degree in Information Systems** | UTN *(In Progress — Expected Graduation: Dec 2027)* · **Bachelor's Studies in Labor Relations & HR** | UBA *(Advanced University Coursework Completed / Organizational Leadership)*.
  - *Certifications* (Verifiable on Credly & Coursera):* Google Data Analytics Professional Certificate | Meta Back-End Developer Certificate | IBM DevOps and Software Engineering Professional Certificate | IBM Full Stack Software Developer (V5) | Oracle Next Education F2 Back-End | Google Cloud Fundamentals: Core Infrastructure.
- **Languages: Spanish:** Native · **English:** Professional Working Proficiency (B2 — Fluent technical communication for live engineering syncs, RFCs, and documentation).